

AHMAD HASAN SAKIB

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RESEARCH INTERESTS

Nonlinear Optics: Waveguide simulations and nonlinear refractive index prediction.
Machine Learning: Predictive modeling in material science, optimization algorithms.
Computational Physics: FDTD methods, data-driven physical modeling.

EDUCATION

Shahjalal University of Science and Technology (SUST) <i>B.Sc. (Hons) in Physics</i>	Sylhet, Bangladesh Jan 2023 – Present
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- **CGPA: 3.80 / 4.00** (Completed up to 3rd Year, 2nd Semester)
- Relevant Coursework: Electromagnetism, Quantum Mechanics, Solid State Physics.

RESEARCH EXPERIENCE

Nonlinear Optics Laboratory, Department of Physics, SUST <i>Undergraduate Researcher</i>	Dec 2024 – Present
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- Implementing **Finite-Difference Time-Domain (FDTD)** simulations for waveguide analysis.
- **Active Project:** *Machine Learning-Driven Prediction of Nonlinear Refractive Indices (n_2) in Multicomponent Glass Systems.*

Computational Climate Science Project <i>Independent Researcher</i>	Feb 2024
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- Processed and analyzed multidimensional **NetCDF** climate datasets to evaluate the correlation between ambient temperature and nuclear power plant efficiency.
- Utilized Python-based libraries (**Xarray**) for spatial-temporal data visualization.

AWARDS & ACHIEVEMENTS

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| • Bronze Medal , University Physics Competition (UPC) | 2025 |
| • Research Portfolio: Developed a Jekyll-based physics blog hosted on GitHub Pages. | |

TECHNICAL SKILLS

- **Languages:** Python (NumPy, Pandas, Scikit-Learn, Xarray),
- **Tools:** Arch Linux, Git/GitHub, VS Code, Markdown, Jekyll.
- **Data Formats:** NetCDF, CSV.

EXTRA-CURRICULAR ACTIVITIES

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| Assistant Cultural Secretary , SUST Physics Society | Jan 2024 – Present |
| • Orchestrating departmental seminars and cultural events for over 300 students. | |